Workload Planning

Laboratory for the Information Systems course
in the degree program
Software Engineering and Media Computing

Milestone 1: Kick Off

submitted by

Team 4

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1 Team Roles

At first the roles were assigned to each team member, followed by the definition of the responsibilities per role. The outcome is described below.

Project Leader and Quality Manager:

Arbresha Selimi

- Organizes and appoints meetings
- Makes sure the schedule and timeline is followed
- Prioritizes requirements, makes sure most important ones are implemented first
- Performs the final review of documents and the project

Design Manager:

Raluca Maria Vedislav

- Coordinates the ER Model
- Creates the conceptual, logical and physical design of the database
- Thinks of the relationships between entities, including constraints and keys

Test Manager:

Pavlos Arampatzis

- Prepares test cases
- Makes sure everything is tested
- Verifies that the program functions properly
- Checks for unexpected situations

Development Manager:

Daniel Ryssel

- Writes code for the database
- Manages the implementation of the design
- Extracts data from PDFs
- Makes sure code follows coding conventions

2 Project Description

In order to understand the project better, a mission statement and the associated mission objectives were discussed in the team.

2.1 Mission Statement

Our mission is to create a system that enables users to efficiently and reliably manage professor workloads and plan their courses in order to replace the current system and improve the overall process.

2.2 Mission Objectives

The following table contains the mission objectives with a short description.

No.	Objective	Description
1	Lecture organization and scheduling	The system should be able to organize lectures by assigning weekly hours for each course. It should allow the same lectures to be utilized across different majors, ensuring efficient use of resources and a standardized, identifiable course ID for tracking purposes.
2	Workload planning	The system should be able to calculate and manage the workload of each professor, including determining the total number of lectures, tutorials and lab sessions conducted weekly. This will ensure a balanced and manageable workload distribution for faculty members.
3	Adaptable scheduling	The system should be able to accommodate changing professor availability and support shared lab sessions among professors, with the ability to split lab responsibilities as needed. With these features, the system will prioritize adaptability and avoid rigid application, ensuring seamless adjustments to changes in teaching hours and staff allocations.
4	Credit hour administration	The system should be able to track and manage credit hours on an individual basis for each professor. It will ensure the accurate recording of teaching contributions, including lectures, tutorials, and labs, for efficient workload management and reporting.
5	Report of lectures, tutorials and labs	The system should be able to create a report listing all lectures, tutorials, and laboratories offered, categorized by degree program and semester. This will provide clear visibility into course offerings, ensuring proper planning and coordination across academic programs.

6	Report of professor workload	The system should be able to create a report providing detailed information for any given professor, including the total weekly hours dedicated to lectures, tutorials, and labs, as well as the current status of their credit hour account. The goal is to smoothen the course and professor management process, reducing errors and simplifying administrative tasks.
7	Report of lecturers	The system should be able to create a report for the Rectorate that includes the planned number of teaching hours per week for each lecturer. The report will also provide lecturer contact details, excluding any storage of professor addresses in the system..
8	Report of provided services	The system should be able to create a report with all services offered to other departments, capturing details such as the subject of the service, the lecturer delivering it, the number of hours per week and the department receiving the service.
9	Report of external services used	The system should be able to create a report for the Rectorate with the services used from one department by another. This will ensure transparent and efficient management of collaborations between departments.

3 Project Timeline

Lastly, a timeline for the project was estimated.

	Stage	Start	End	2024				2025
				Sep	Oct	Nov	Dec	Jan
	Project	24.09.24	07.01.25					
1	Planning	24.09.24	01.10.24					
2	System Analysis	01.10.24	15.10.24					
3	System Design	15.10.24	29.10.24					
4	Implementation	29.10.24	12.11.24					
5	Testing	12.11.24	03.12.24					
6	System Acceptance	03.12.24	07.01.25					

Following tasks are planned for each stage:

- **Planning:** Database planning, system definition
- **System Analysis:** Requirements collection and analysis
- **System Design:** Database design, DBMS selection, application design
- **Implementation:** Implementation, Data conversion and loading
- **Testing:** Testing the system, fixing mistakes, final tweaks
- **System Acceptance:** Delivery of final product and evaluation